

	Course Name:	Engineering Management	Course Code:	MG 330
	Program:	BS(Electrical Engineering)	Semester:	Fall 2018
	Duration:	150 Minutes	Total Marks:	100
	Paper Date:	19-December-18	Weight	50%
	Section:	ALL	Page(s):	4
	Exam Type:	Final		

Student : Name: [REDACTED] Roll No. [REDACTED] Section: EE-1

Instruction/Notes: Start each question from new page, show all steps, and assume if something is missing in paper.
Use Products

Q #1:

- (a) Two element of a business's functional structure are "Customers and Suppliers". Name at least six more elements of modern-day engineering organization functional structure (6 marks) [CLO-5]
- (b) Draw a block diagram and show how these elements of modern day functional structure in part (a) interact with each other using directional arrows. [10 marks] [CLO-5]

Q #2:

- (a) Name five typical stakeholders in any business. Describes expectations of each of these stakeholders. [10 marks] [CLO-4]
- (b) One of the key responsibilities of a manger is to make decisions about planning, organizing, leading and controlling. Pakistan (and many other countries are argued to be having economies that are poorly managed. Discuss each of planning, organizing, leading and controlling in terms of their contribution to the poor management of the economies of these countries. Draw a conclusion wherein you list all of planning, organizing, leading and controlling by ranking them for their impact as highest to the lowest on poor management of the economies. [10 marks, Not to exceed 300 words] [CLO-1,4-7]

Q #3: Answer the following multiple choice questions by encircling the correct answer on the question paper. You must submit this question paper along with the answer sheet. [2 marks per MCQ, Total 30 marks]

- Which of the following is not a controlling activity? [CLO-7]
 - assess and regulate work in progress
 - evaluate results for securing and maintaining maximum productivity
 - reduce and prevent unacceptable performance
 - ☒ Suitable arrangement of the interrelated groups of people participating in the work
- Which of the following statement is not true? [CLO-1]
 - Companies need effective managers to optimize profits
 - Planning is the work done to predetermine a course of action
 - ☒ Planning defines who will do what, how, where, when, and with which resources
 - Organizing means arranging and relating work so that it can be done efficiently
- Find the odd one out? [CLO-6]
 - A leader sets a good personal example
 - A leader is able to attract and retain good people
 - ☒ A leader induces people to unwillingly do their best
 - A leader motivates people to use their abilities

12
30

4. What business analysis is not about? [CLO-4]
 - a. opportunities identification
 - b. threats identification
 - c. strength identification
 - d. None of the above
5. Making suitable arrangement of the interrelated groups of people participating in the engineering management aspect? [CLO-5]
 - a. Planning
 - b. Leading
 - c. organizing
 - d. Controlling
6. Which of the following is relevant to HR performance review? [CLO-7]
 - a. Educational credentials
 - b. Communication skills
 - c. domicile
 - d. Marital status
7. Pick that relates to soft skills? [CLO-6]
 - a. Technical expertise
 - b. Position in the company
 - c. Leadership
 - d. Collaborative attitude
8. Which style of leadership is not realistic? [CLO-6]
 - a. Placing too much value on social acceptance while neglecting technical tasks
 - b. Balancing both the needs of people and task
 - c. Focusing on getting the job done right without concern for human feelings
 - d. Getting the job done and at the same time making everyone happy
9. Which organizational structure is most suited for a self-driving car start-up? [CLO-5]
 - a. Functional
 - b. Region based
 - c. Matrix
 - d. Discipline based
10. Which of the following is not a consequence of delegating? [CLO-5]
 - a. Improvement in quality
 - b. getting important duties done efficiently
 - c. employees salary enhancement
 - d. Distribution of work load efficiently
11. Which of the following statement is not correct? [CLO-5]
 - a. Conflicts may be resolved by Dominance
 - b. Conflicts may be resolved by Compromise
 - c. Conflicts may be resolved by Collaboration
 - d. Conflicts may be resolved by Requesting
12. Which of the following is not an action planning step? [CLO-4]
 - a. Analyze critical needs
 - b. Develop a human resource
 - c. Define key action steps
 - d. Create a schedule
13. Which of the following relates to an engineering manager necessarily? [CLO-1]
 - a. Demonstrating best possible technical skills in performing tasks and projects
 - b. Practicing engineering principles correctly
 - c. Demonstrating business savviness
 - d. Demonstrating innovative capabilities in product design
14. Which of the following statement is wrong? [CLO-4,5,6,7]
 - a. Span of control is the number of people managed by a supervisor
 - b. Inspiring is a method of employee selection
 - c. Engineering managers secure result by motivating people

- d. Specialization is the increased degree of skill in narrow technical domain
15. Which of the following is not a corporate intellectual property? [CLO-7]
- patents
 - proprietary know-how
 - operational duty roster
 - empirical problem-solving heuristics

$$P = F$$

$$F = \frac{P}{(1+i)^n}$$

Q#4: Draw a work breakdown structure for the construction of house with a level of 5*3 (5 horizontal, 3 vertical) (7 marks) [CLO-2]

Q#5: A metallurgical engineer is considering two materials for use in space vehicle. On the basis of net present value compare the given materials at an interest rate of 12% per year. (Formulas are not given). (7 marks) [CLO-8]

	Material X	Material Y
First Cost, \$	15,000	35,000
Maintenance Cost, \$ per year	9,000	7,000
Salvage value, \$	2,000	20,000
Life, years	5	5

$$F = P(1+i)^n$$

Q#6: The Global Tea and Organic juice companies have merged. The following information has been collected for the "Consolidation Project." (show all calculations). (20 marks) [CLO-3]

Activity	Description	Predecessor	a	m	b
1	Codify accounts	-	16	19	28
2	File articles of unification	-	30	30	30
3	Unify price and credit policy	-	60	72	90
4	Unify personal policies	-	18	27	30
5	Unify data processing	1	17	29	47
6	Train accounting staff	1	4	7	10
7	Pilot run data processing	5	12	15	18
8	Calculate P & L and balance sheet	6,7	6	12	24
9	Transfer real property	2	18	27	30
10	Train Salesforce	3	20	35	50
11	Negotiate with unions	4	40	55	100
12	Determine capital needs	8	11	20	29
13	Explain personal policies	11	14	23	26
14	Secure line of credit	9,12	13	16	19
15	End	10,13,14	0	0	0

- Compute the expected time for each activity.
- Compute the variance for each activity.
- Draw a neat PERT diagram with critical path by using the box given below (you can ignore description).
- Compute the expected project duration.
- What is the probability of completing the project by day 112? Within 116 days? 114?

ES	ID	EF
SL	Description	
LS	DUR	LF

Q#06:

Part (a) and (b)

Rough Work

Activity	a	m	b	te	variance for each activity
✓ 1	16	19	28	20	$(\frac{b-a}{6})^2$
2	30	30	30	30	$(2)^2 = 4$
3	60	72	90 90	73	0
✓ 4	18	27	30 30	26	$(5)^2 = 25$
✓ 5	17	29	47 47	30	$(2)^2 = 4$
6	4	7	10 10	7	$(5)^2 = 25$
✓ 7	12	15	18	15	1
✓ 8	6	12	24	13	01
9	18	27	30	26	9
10	20	35	50	35	4
11	40	55	100	60	$(5)^2 = 25$
✓ 12	11	20	29	20	$(10)^2 = 100$
13	14	23	26	22	$(3)^2 = 9$
✓ 14	13	16	19	16	$(2)^2 = 4$
✓ 15	0	0	0	0	1
					0

$$te = \frac{(a + b + 4m)}{6}$$

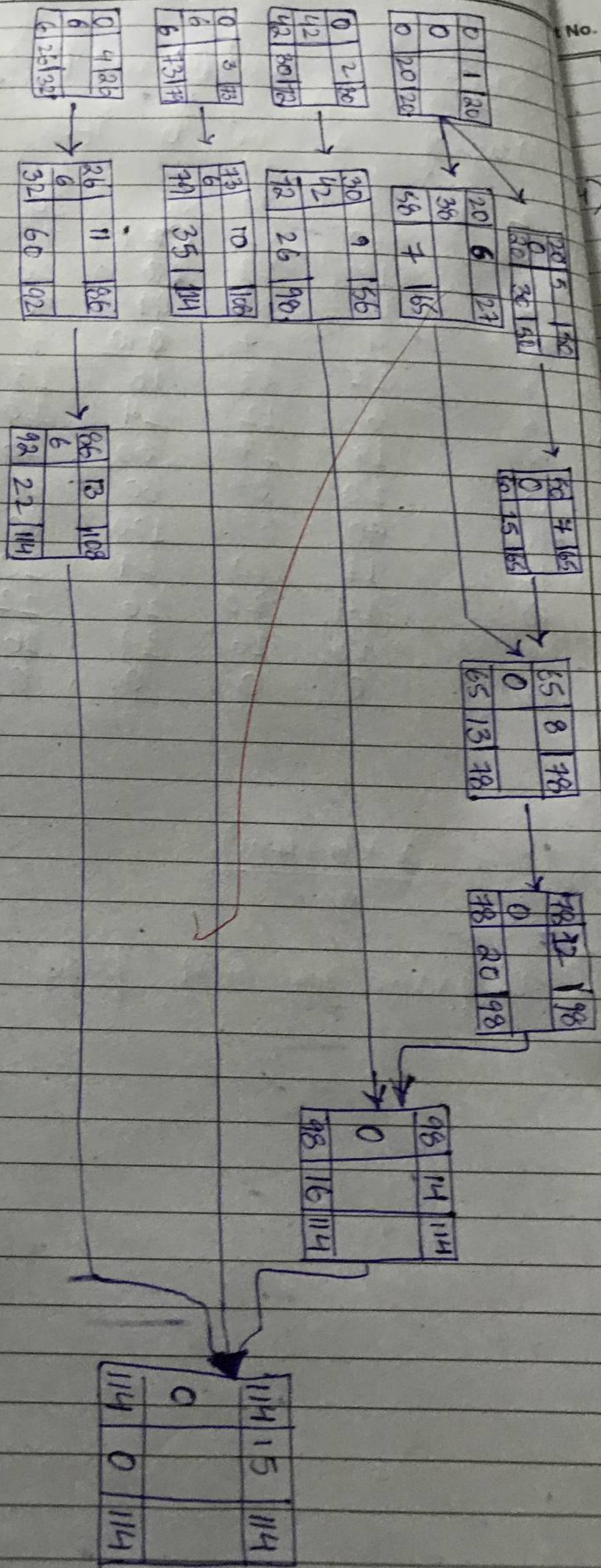
Diagram on next page

Q7 Part No. 3

part(3)

CRITICAL PATH:

1-5-7-8-12-14-15



Q#ct (cont)

4) Expected project duration:

$$20 + 30 + 16 + 0 + 20 + 13 + 15 \\ = 114 \text{ (days)}$$

(5) (a) Probability for 112 days:

Let $T_{s1} = 112$ days.

$$Z = \frac{T_{s1} - T_E}{\sqrt{\sum (\sigma_{LE})^2}}$$

$$\sum (\sigma_{LE})^2 \checkmark \text{ variance of project:} \\ = 4 + 25 + 1 + 9 + 9 + 1 + 0 = 49$$

$$Z = \frac{112 - 114}{\sqrt{49}} = -0.285$$

from given table:

$$P = 0.3859$$

Q / Part No.

Q#6(s) b Probability for 116 days.

$$\text{let } T_{52} = 116$$

$$\Rightarrow Z = \frac{116 - 114}{\sqrt{49}} \\ = 0.285$$

$$P = 0.6141$$